

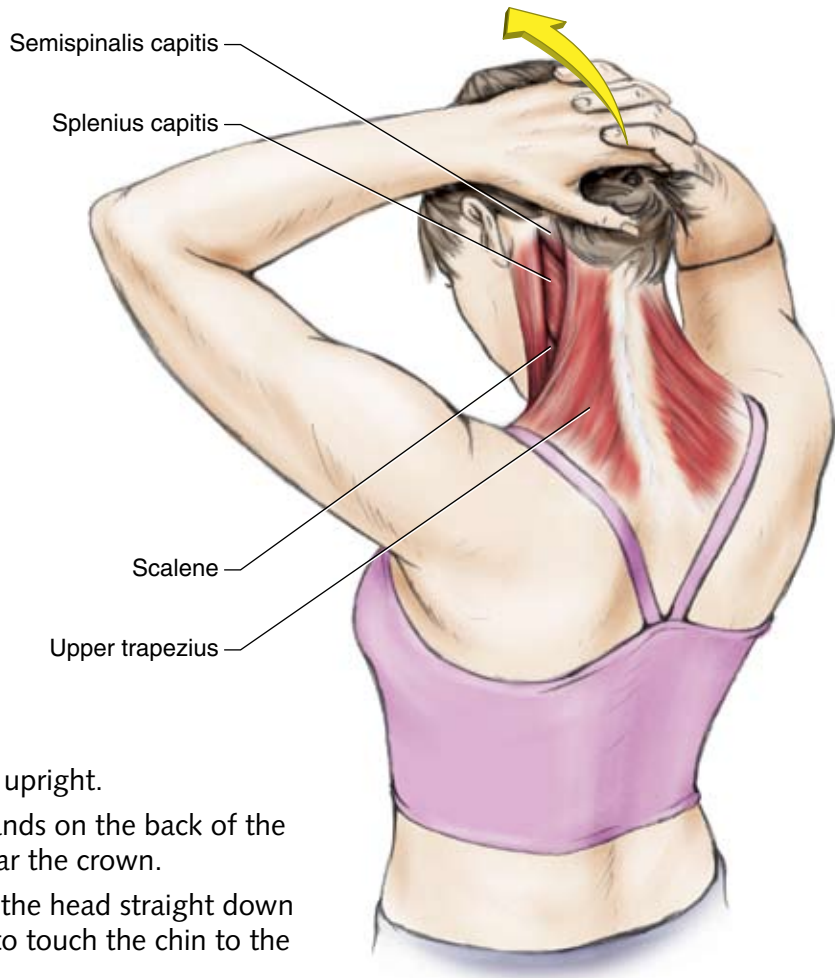
In the neck, the muscles are located in two triangular regions called the anterior and posterior triangles. The borders of the anterior triangle are the mandible (jawbone), the sternum (breast bone), and the sternocleidomastoid muscle. The major anterior muscles are the sternocleidomastoid and scalene. The borders of the posterior triangle are the clavicle (collarbone), sternocleidomastoid muscle, and trapezius muscle. The major posterior muscles are the trapezius, longissimus capitis, semispinalis capitis, and splenius capitis. The muscles in the neck are involved primarily in supporting or moving the head. The head movements are flexion (head tilted forward), extension (head tilted backward), lateral flexion and extension (head up and back sideward), and rotation. Since the muscles in the neck come in right and left pairings, all of the neck muscles are involved with lateral flexion and extension. For example, the right sternocleidomastoid helps perform right lateral flexion, and the left sternocleidomastoid helps perform right lateral extension. Illustrations showing the muscles and movements as well as a chart showing which muscles do a specific movement are located at the end of the chapter (pages 6-7).

When people think about doing stretching exercises, they seldom consider the neck muscles. Neck flexibility probably does not cross your mind until you discover that you have a stiff neck. A stiff neck is commonly associated with sleeping in a strange position (such as on a long flight), but a stiff neck can result from almost any type of physical activity. This is especially true for any activity where the head must be held in a constant stable position. Thus, a stiff neck can also have a negative effect in sports where head position is important (such as golf) or where rapid head movements are important for tracking the flight of an object (such as in racket sports). Poor neck flexibility usually results from holding the head in the same position for long periods. In addition, a fatigued neck muscle can stiffen up after exercise. The following exercises can help keep the neck from stiffening up after exercise, unusual postures, or awkward sleep positions.

Since all of the major muscles in the neck are involved in neck rotation, it is fairly easy to stretch the neck muscles. The first consideration when choosing a particular neck stretch should be whether greater stiffness occurs with flexion or extension. Therefore, the first two exercise groups are concerned with these specific actions. Once you achieve greater flexibility in either pure flexion or pure extension, then you can add a stretch that includes lateral movement. In other words, to increase the flexibility of the neck extensors, start with the neck extensor stretch and then, as flexibility increases, add the neck extensor and rotation stretch.

Remember that overstretching (very hard stretching) causes more harm than good. Sometimes a muscle becomes stiff from overstretching. Stretching can reduce muscle tone, and when tone is reduced, the body compensates by making the muscle even tighter. For each progression, start with the position that is the least stiff and progress only when, after several days of stretching, you notice a consistent lack of stiffness during the exercise. This means that you should stretch both the agonist muscles (the muscles that cause a movement) and antagonist muscles (the muscles that oppose a movement or do the opposite movement). Also, remember that although you may have greater stiffness in one direction (right versus left), you need to stretch both sides so that you maintain proper muscle balance.

Neck Extensor Stretch



Technique

- Sit or stand upright.
- Interlock hands on the back of the head near the crown.
- Lightly pull the head straight down and try to touch the chin to the chest.

Muscles Stretched

Most-stretched muscle: Upper trapezius.

Lesser-stretched muscles: Longissimus capitis, semispinalis capitis, splenius capitis, scalene.

Commentary

You can do this stretch either while sitting or while standing. A greater stretch is applied when seated. Standing reduces the ability to stretch because reflexes come into play to prevent a loss of balance. During the stretch, make sure not to reduce the stretch by hunching up the shoulders. Also, keep the neck as straight as possible (no curving). Try to touch the chin to the lowest possible point on the chest.

VARIATION

Neck Extensor and Rotation Stretch

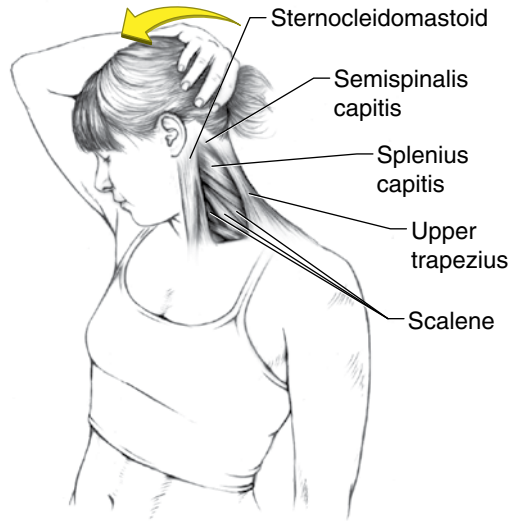
When the neck extensors become flexible, progress from stretching the right and left sides simultaneously to stretching the opposite sides individually. To do this, follow this procedure:

Technique

Sit or stand upright.

Place the right hand on the back of the head near the crown.

Pull the head down toward the right and try to touch the chin as close as possible to the right shoulder.

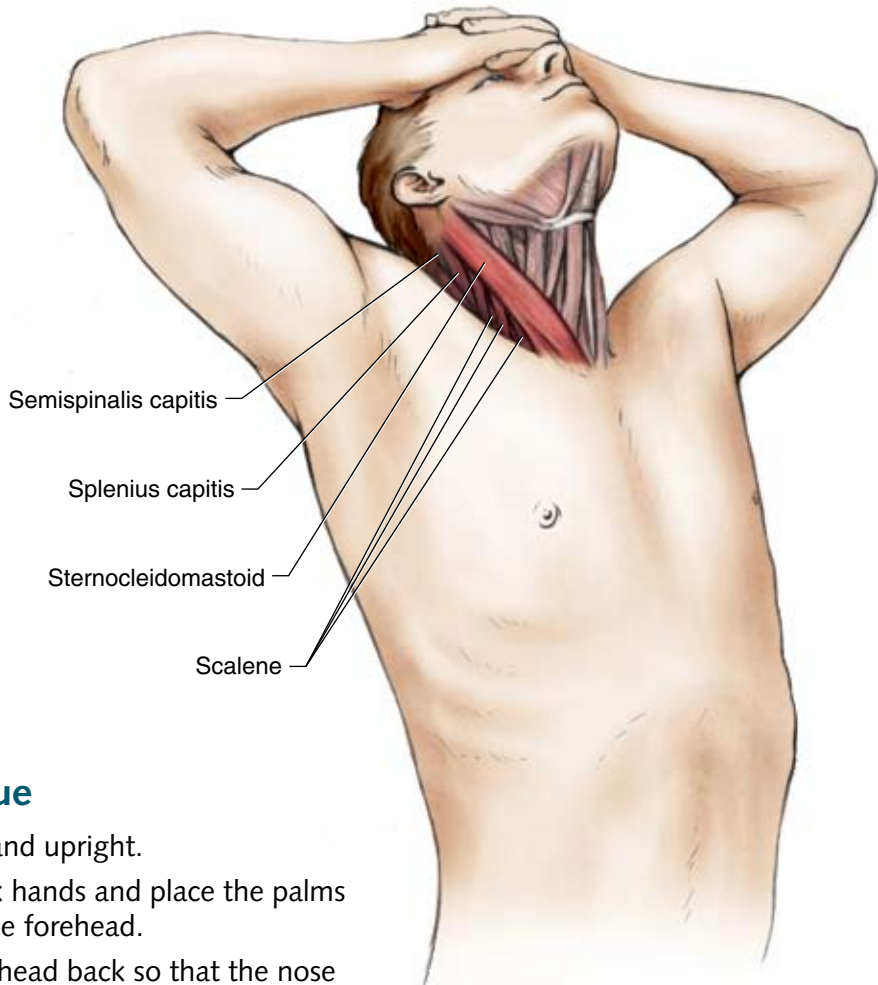


Muscles Stretched

Most-stretched muscles: Left upper trapezius, left sternocleidomastoid.

Lesser-stretched muscles: Left longissimus capitis, left semispinalis capitis, left splenius capitis, left scalene.

Neck Flexor Stretch



Technique

- Sit or stand upright.
- Interlock hands and place the palms on the forehead.
- Pull the head back so that the nose points straight up to the ceiling.

Muscles Stretched

Most-stretched muscle: Sternocleidomastoid.

Lesser-stretched muscles: Longissimus capitis, semispinalis capitis, splenius capitis, scalene.

Commentary

You can do this stretch either while sitting or while standing. A greater stretch is applied when seated. Standing reduces the ability to stretch because reflexes come into play to prevent a loss of balance. During the stretch, make sure not to reduce the stretch by hunching up the shoulders. Also, try to point the chin as far back as possible.

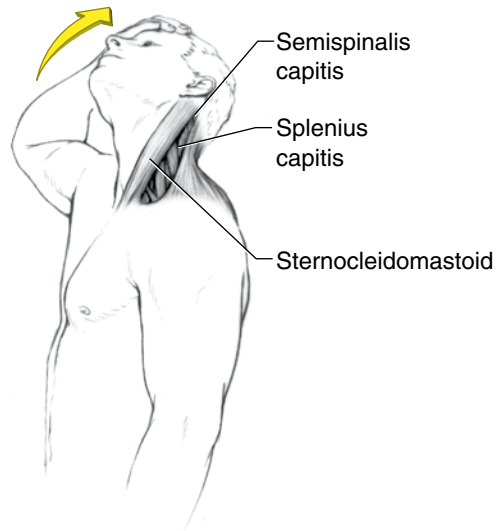
VARIATION

Neck Flexor and Rotation Stretch

When the neck flexors become flexible, progress from stretching the right and left sides simultaneously to stretching the opposite sides individually. To do this, follow this procedure:

Technique

- Stand or sit upright.
- Place right hand on the forehead.
- Pull the head back and toward the right so that the head points toward the shoulder.
- Keep the head straight; do not lay the head down to either side.



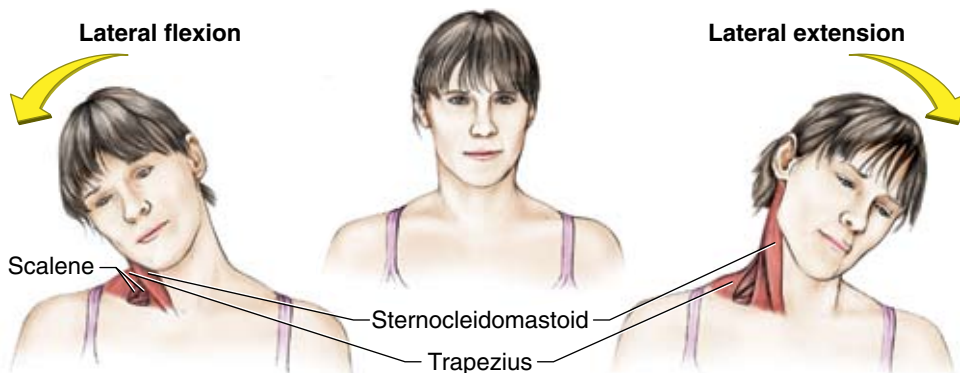
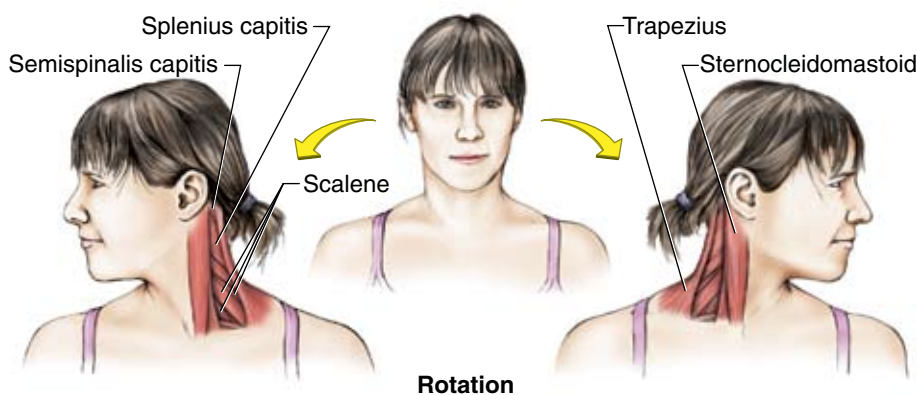
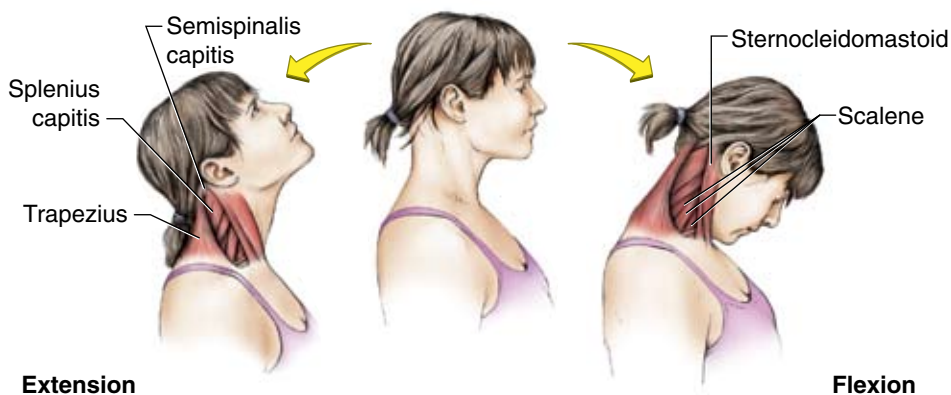
Muscles Stretched

Most-stretched muscle: Left sternocleidomastoid.

Lesser-stretched muscles: Left longissimus capitis, left semispinalis capitis, left splenius capitis.

Neck Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various neck movements appear in the following table. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the left scalene, you could extend the head both backward and laterally to the right. When a muscle has a high level of stiffness, you should use fewer simultaneous opposite movements (you would stretch a very tight right scalene by initially doing just left lateral extension). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.



Muscle	Flexion	Extension	Rotation	Lateral flexion	Lateral extension
Longissimus capitis		✓	✓	✓	✓
Scalene	✓		✓	✓	
Semispinalis capitis		✓	✓	✓	✓
Splenius capitis		✓	✓	✓	✓
Sternocleidomastoid	✓		✓	✓	✓
Trapezius		✓	✓	✓	✓